

IO Research

Cardano Vision 2026

Human Centred, Scalable, and Post Quantum Secure

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Who Should Read This Document

This Cardano Vision 2026 proposal is intended for all stakeholders involved in shaping the future of Cardano, including Delegated Representatives (DReps), the Constitutional Committee, Stake Pool Operators (SPOs), builders, and the wider community. It provides a clear and structured view of the technical program, enabling informed participation in governance, funding decisions, and ecosystem development, and supporting the collaborative, decentralized decision-making model that underpins Cardano's evolution.

1. Technical Program

The Cardano Vision 2025 proposal was approved with strong support (4.16B ada in favour, 79.81%), reflecting clear confidence in the role of research in Cardano's long-term strategy. Cardano Vision launched as a five-year initiative grounded in peer-reviewed science to secure Cardano's technical leadership.

In 2025, IOR delivered against this mandate, **advancing 20 research streams** and **publishing 24 peer-reviewed papers**, exceeding targets by **20%**. More importantly, this work translated into tangible protocol progress. In consensus, Ouroboros Peras and Leios demonstrated fast settlement and scalable throughput, with formal proofs and a Crypto '25 publication, alongside advances in fair transaction ordering. In interoperability, the first formal bridge security framework and the Cavefish light client were introduced. In tokenomics, new reserve and pooling models were developed, including Shapley-based reward mechanisms. In zero-knowledge, frameworks such as AGATE and UC-SNARKs advanced Cardano's position in verifiable computation.

Alongside research, **eight technology validation streams** moved key innovations toward deployment. Leios progressed to prototype and was handed over to engineering with a CIP; anti-grinding protections (Phalanx) were formalised; Jolteon achieved prototype readiness; and ZK verification (RSnarks) demonstrated on-chain feasibility. These outputs show a clear trajectory from theory to implementation.

This progress establishes a strong foundation, but also highlights the next challenge: improving the consistency, speed, and impact of research translation. Since its inception, IOR has produced over 250 peer-reviewed papers, with approximately 20% implemented in Cardano – demonstrating both the strength of the research program and the opportunity to increase conversion into deployed systems.

Cardano Vision 2026 focuses on **scaling this impact** through a more structured, **product-aligned delivery model**, ensuring that research outputs translate more reliably into measurable ecosystem growth.

IO Research: An Integrated Delivery Model

IOR implements an integrated operating model to combine research (IOR), applied engineering (ARC), and product (CBU), into a single delivery pipeline.

- **Input Output Research (IOR):** scientific investigation, new protocols, models, and security proofs
- **Applied Research & Creative Engineering (ARC):** validates feasibility through prototypes, simulations, and specifications
- **Cardano Business Unit (CBU):** ensures alignment with real-world use cases, adoption, and go-to-market

This structure replaces a linear research process with a **continuous innovation pipeline**, where work is aligned to product outcomes and progressively de-risked. Research is

embedded within a continuous pipeline linking discovery to application, with aligned incentives and accountability across functions. IOR operates as a staged funnel, as follows:

- **Fundamental Research ($\leq$ TRL2):** new protocol designs and formal analysis
- **Technology Validation (TRL3–5):** prototypes, simulations, and specifications
- **Implementation (TRL5+):** engineering support for production-ready systems

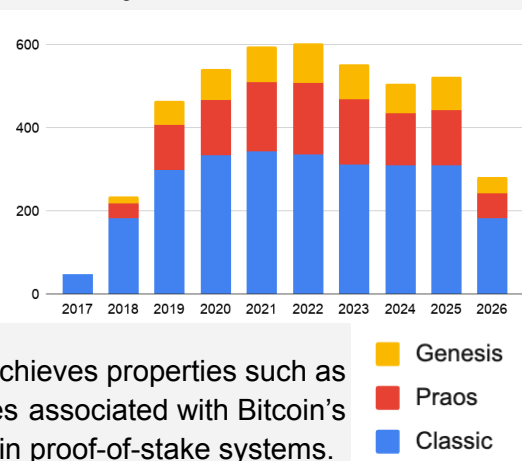
The result is a disciplined innovation engine, preserving Cardano's evidence-based approach while ensuring **workstreams are grounded in product relevance** and adoption pathways to deliver impact for the community.

Illustrative Use Case: Ouroboros

Ouroboros is the foundation of the Cardano ledger, delivering consensus through a longest-chain proof-of-stake (PoS) protocol that governs block production and validation.

It represents one of the most rigorously developed consensus systems in blockchain, combining formal security guarantees with practical deployment at scale. The protocol family has been cited over 4,000 times and has influenced or been adopted by other major systems, including Polkadot, Mina, and Horizen.

Figure 1: Ouroboros citations over time



A defining characteristic of Ouroboros is its emphasis on decentralization and resilience. It achieves properties such as dynamic availability and self-healing – capabilities associated with Bitcoin's proof-of-work model, but not typically achievable in proof-of-stake systems.

The Long Arc of Research

Since its initial deployment in 2017, Ouroboros has evolved through a series of iterations, each one strengthening security, scalability, and operational robustness.

- **Classic:** established provably secure longest-chain PoS with unbiased, on-chain randomness for leader selection
- **Praos (production):** introduced private leader selection and forward-secure key evolution to protect against adaptive attacks
- **Genesis (in development):** enables secure bootstrapping from genesis without trusted checkpoints under dynamic participation
- **Omega (research):** the intended 'final form' combining adaptive performance with consistent security

A core innovation across the Ouroboros family is the use of unbiased randomness for leader selection, providing strong, formally verified guarantees. Each iteration builds

incrementally, extending the protocol's ability to operate securely under real-world conditions.

Omega advances this further by combining optimistic execution – scaling performance when conditions allow, falling back safely when they do not – with multi-resource security, extending protection beyond stake alone. The result is a protocol where security remains constant while performance adapts dynamically.

Technology Innovation

In parallel, IOR supports targeted upgrades to enhance Praos performance:

- **Peras:** reduces settlement latency, improving transaction finality and user experience
- **Leios:** increases throughput by optimizing block propagation and network efficiency

These upgrades are modular, allowing specific performance characteristics to be improved without compromising core security guarantees.

Implementation Pathway

Peras and Leios are expected to reach deployment readiness in 2026. They illustrate how research progresses through the IOR pipeline, from formal design to validation and integration, before being delivered to production environments.

IOR continues to support this process through testing, benchmarking, and deployment readiness, ensuring that protocol improvements are both theoretically sound and operationally robust (see Consensus below).

Scope and Approach

The technical program for Cardano Vision 2026 is designed to translate research into deployable system capabilities. It integrates foundational research, engineering validation, and product alignment into a set of work packages that advance the network toward its 2030 objectives. Rather than operating in isolation, each workstream targets concrete system properties – security, performance, usability, and economic sustainability – ensuring direct contribution to measurable ecosystem outcomes.

Cross-Cutting Strategic Themes

Three themes underpin Cardano Vision 2026: post-quantum security, scalable architecture, and human-centred design. These themes underpin all work packages.

- **Human-centred design** focuses on usability, incentives, and governance participation. This includes improving developer experience, strengthening economic alignment for SPOs, and reducing friction for users through Intents.
- **Scalability** is addressed through a layered approach, combining consensus improvements (eg, Leios, Peras), execution scaling (eg, L2s, data availability), ZK-enabled scaling (eg, ZK rollups), and longer-term research (eg, sharding), delivering increased capacity without compromising decentralization.
- **Post-quantum security** ensures long-term resilience as cryptographic assumptions evolve, with proactive evaluation and migration planning for quantum-resistant primitives.

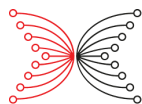
The Cardano Vision 2026 proposal is organised into **six clusters**, each directly linked to measurable ecosystem outcomes:

- **Trust, Security & Reliability Infrastructure**: enterprise-grade resilience and continuous availability of the base layer
- **Scalability & Execution Layer**: increased throughput and execution capacity across L1 and L2 while preserving settlement guarantees
- **Developer Platform & User Experience**: reduced friction for builders and users, supporting adoption and MAU growth
- **Applications, Adoption, & Liquidity**: increased on-chain demand, capital inflows, and real-world usage
- **Economic Systems & Incentives**: aligned, sustainable economics across treasury, validators, and protocol revenue
- **Governance, Identity, & Social Infrastructure**: robust coordination systems, decentralized governance, and identity primitives

Together, these clusters define a structured pathway from research to deployment where each work package solves a technical problem and delivers capabilities that strengthen and enhance Cardano.

WP1: Trust, Security, & Reliability Infrastructure

Leads	William Wolf, Varun Maram, Mario Larangeira
FTE	8.3 (5 IOR, 4.6 ARC)
KPIs	1.1 Consensus: Adoption, Scalability, & Reliability • <i>Monthly Transactions</i>: Enables growth towards 27M monthly transactions • <i>Throughput capacity per day</i>: Supports 3× scaling target • <i>Monthly Uptime</i>: Maintains continuous block production under load



	1.2 Post-Quantum Security: Reliability & Sustainability • <i>Monthly Uptime</i>: Ensures long-term protocol stability • <i>Annual Protocol Revenue</i>: Protects long-term economic activity and capital 1.3 Node Security: Reliability & Resilience • <i>Monthly Uptime</i>: Reduces risk of validator failure • <i>Operational Resilience</i>: Supports decentralization and reduces concentration risk
Pillars	P1: Infrastructure & Research Excellence 1.1 Consensus • <i>1.1 Scalability & Interoperability, L1 Protocol improvements</i>: Improving consensus, transaction ordering, throughput, and finality (Leios, Peras) • <i>1.2 Security & Resilience, Threat detection & recovery</i>: Adversarial modelling, mempool attacks, and defining safe operational boundaries 1.2 Post-Quantum Security • <i>1.2 Security & Resilience, Post-quantum readiness</i>: Migration of signatures, VRFs, and protocol cryptography • <i>1.2 Security & Resilience, Threat detection & recovery</i>: Ensuring long-term protocol survivability under new adversarial models 1.3 Node Security • <i>1.2 Security & Resilience, Threat detection & recovery</i>: Secure key management, reduced attack surface, improved operational resilience • <i>1.2 Security & Resilience, Client diversity</i>: Strengthening node infrastructure and operational robustness

WP1 Trust, Security, & Reliability Infrastructure delivers formally-verified and optimized consensus improvements, alongside robust node infrastructure and post-quantum readiness. These factors ensure continuous operation and enable scalable growth under real-world adversarial conditions.

- Consensus assurance package including MEV & mempool attack analysis, adversarial modelling of Leios, and protocol optimisation and integration studies to define secure operating boundaries. (TRL 6–8)

- Produces a post-quantum cryptography baseline and evaluation suite and novel PQ VRF construction with formal proof to establish future-ready primitives. (TRL 1–2 > TRL 3–4)
- Delivers quantum-secure Ouroboros analysis, integration specifications, and performance benchmarks to enable implementation-ready post-quantum infrastructure. (TRL 1 > TRL 4)
- Develops node security architecture including threat models and HSM-backed key management prototypes to strengthen operational resilience. (TRL 2 > TRL 4)

1.1 Consensus

This workstream focuses on evolving Cardano’s consensus layer to deliver secure, efficient, predictable, and economically robust transaction execution, while preserving the strong guarantees of the Ouroboros protocol. Rather than prioritizing performance alone, it treats transaction ordering, consistent level of service, and validator incentives as first-class design constraints – reflecting their importance for real-world financial applications.

(i) MEV, Mempool Attacks & Safety Analysis

A central requirement is predictable transaction ordering under adversarial conditions. This work analyses maximal extractable value (MEV) within the EUTXO model, focusing on Linear Leios as the next-generation throughput upgrade. It examines MEV vectors such as mempool partitioning and ordering manipulation, and evaluates whether EUTXO’s structural properties constrain extractable value relative to account-based systems.

(ii) Protocol Validation & Operational Boundaries

The protocol must operate within clearly defined performance and security limits. This requires validating Linear Leios under both adversarial and degraded conditions, modelling behaviour across optimistic and fallback execution modes, and analysing mempool consistency. A key challenge is balancing performance gains with decentralization, as higher throughput may introduce non-linear resource demands on operators. This work defines safe operational envelopes – covering validator requirements, resource constraints, and failure modes – to ensure predictable and accessible network operation.

(iii) Targeted Protocol Improvements (Leios & Peras)

Targeted improvements aim to improve efficiency without weakening security guarantees. This includes reducing the likelihood of observing cooldowns in Peras, optimizing voting certificate construction for both Peras and Leios to balance communication and on-chain storage costs, and analysing the trade-offs between throughput and resource efficiency. The interaction between Leios and Peras is explored to exploit synergies of joint deployment.

Out of scope: This scope excludes the productionisation of Linear Leios in the proposed IO: Consensus Initiative including hard-fork preparation, parameter validation and graduation,

large-scale testnet and adversarial testing, red-teaming, and production threat modelling.

1.2 Post-Quantum Security

Cardano's security model relies on cryptographic primitives, digital signatures, and verifiable random functions (VRFs), whose current instantiations (eg, Ed25519, ECVRF) are vulnerable to quantum attacks. While large-scale quantum computers do not yet exist, advances in both algorithms and hardware are rapidly reducing the threshold for practical attacks. The objective of this stream is therefore to establish a credible, evidence-based pathway for migrating Cardano to post-quantum security.

(i) Post-Quantum Cryptographic Primitives

The first requirement is the identification and development of quantum-secure replacements for core protocol components, in particular digital signatures, VRFs, and threshold schemes. Among these, VRFs present the most significant challenge due to their role in leader election and the practical constraints on proof size and verification time. The work therefore focuses on evaluating candidate constructions and developing post-quantum VRFs that meet Cardano's performance and composability requirements.

(ii) Protocol Integration and Formal Security in a Quantum Setting

Replacing cryptographic primitives in isolation is insufficient; the security of the Ouroboros protocol must be re-established under quantum adversarial models. This requires adapting existing formal analyses to the post-quantum setting and proving that the upgraded protocol maintains its security guarantees under quantum attack assumptions.

(iii) Migration Strategy and Feasibility Assessment

A central challenge is that full replacement of cryptographic primitives may introduce unacceptable performance or compatibility trade-offs. The work therefore evaluates alternative migration strategies, including approaches that minimise disruption (eg, address migration schemes), and assesses their applicability to Cardano. This includes benchmarking candidate primitives and prototypes early in the process to ensure theoretical work remains grounded in practical feasibility.

(iv) Cross-Cutting Efforts

Post-quantum security is a system-wide concern and therefore extends across multiple workstreams including:

- *Node Security (WP1.3)*: assessing HSM providers and key management architectures capable of supporting post-quantum primitives (eg, VRFs, KES)
- *Congestion Control & Transaction Efficiency (WP2.1)*: evaluating the impact of increased computational overhead on network performance and identifying resource allocation strategies

- *Layer 2 & ZK Verification Infrastructure (WP2.4)*: assessing quantum-secure proof systems (eg, lattice- and hash-based) and their integration into Cardano’s verification stack
- *Bridges, Atomic Swaps & Agent-Based Interoperability (WP4.1)*: exploring post-quantum threshold signatures and their application to cross-chain coordination and external system interaction

1.3 Node Security

The security of the Cardano network depends on the protection of operational cryptographic keys – KES signatures, VRFs, and related credentials – used by stake pool operators for block production, delegation, and potentially governance. These keys vary in criticality, but all represent sensitive assets whose compromise could impact node integrity or protocol security.

Today, key management is largely software-based, exposing keys to risks such as host compromise, memory inspection, and operational error. Strengthening key management is therefore essential to maturing Cardano as production-grade infrastructure. This work begins with a technical assessment of node key architecture and threat models, analysing the role and sensitivity of each key type, complemented by SPO consultation to capture operational requirements. The outcome is expected to inform two complementary approaches:

(a) Hardware-Backed Key Management

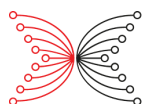
Integration of hardware security modules (HSMs) into the Cardano node to ensure KES and VRF keys are generated, stored, and used within tamper-resistant environments. This extends existing infrastructure (eg, KES agent) via standard interfaces such as PKCS#11, while maintaining block production performance. The work includes validation of key lifecycle processes and delivers a production-ready implementation, benchmarks, and SPO deployment guidance.

(b) Cryptographic Key Management Approaches

Exploration of software-based alternatives such as MPC and threshold cryptography, enabling distributed key control without dedicated hardware. This includes evaluation of hot/cold architectures and forward-looking options such as post-quantum key management, with the aim of supporting flexible, secure configurations aligned to different threat models and operator needs.

WP2: Scalability & Execution Layer

Leads	Giorgos Panagiotakos, Matthias Fitzl, Jesus Diaz Vico
FTE	9.9 (4.5 IOR, 7.4 ARC)



KPIs	2.1 Congestion Control & Transaction Efficiency: Adoption, Scalability & Revenue • <i>Monthly Transactions</i>: Improves transaction throughput and reduces friction under load • <i>Throughput capacity per day</i>: Supports scaling toward 3× capacity target • <i>Annual Protocol Revenue</i>: Optimises fee mechanisms and transaction efficiency 2.2 Sharding: Scalability • <i>Throughput capacity per day</i>: Explores long-term horizontal scaling potential • <i>Monthly Transactions</i>: Enables future step-change in transaction volume 2.3 Communication Infrastructure: Adoption • <i>Monthly Active Users</i>: Reduces UX friction via intent-based interactions • <i>Monthly Transactions</i>: Enables new transaction flows (eg, solver-based execution) 2.4 Layer 2 & ZK Verification Infrastructure: Adoption, Scalability • <i>Monthly Transactions</i>: Enables high-frequency L2 activity anchored to L1 • <i>Total Value Locked</i>: Supports DeFi, bridging, and capital-efficient settlement • <i>Throughput capacity per day</i>: Offloads execution while preserving L1 integrity
Pillars	P1: Infrastructure & Research Excellence 2.1 Congestion Control & Transaction Efficiency • <i>I.1 Scalability & Interoperability, L1 protocol improvements</i>: Improving transaction efficiency, fee mechanisms, and data availability for L2s • <i>I.1 Scalability & Interoperability, L2 integration</i>: Supporting L2 requirements through improved on-chain data availability 2.2 Sharding • <i>I.1 Scalability & Interoperability, L1 protocol improvements</i>: Exploring state partitioning and execution scalability at the base layer • <i>I.1 Scalability & Interoperability, L2 integration (comparison)</i>: Evaluating trade-offs vs rollups and DA-based scaling approaches

	2.3 Communication Infrastructure • <i>1.1 Scalability & Interoperability – L2 integration:</i> Enabling coordination between users, solvers, and applications • <i>1.1 Scalability & Interoperability – Cross-chain interoperability:</i> Supporting messaging and coordination across systems 2.4 Layer 2 & ZK Verification Infrastructure • <i>1.1 Scalability & Interoperability, L2 integration:</i> Enabling scalable execution via L2 settlement anchored to L1 • <i>1.1 Scalability & Interoperability, Core ZK capabilities:</i> Providing verification primitives for ZK-based computation and settlement
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WP2 Scalability & Execution Layer drives step-change improvements in throughput and execution capacity across L1 and L2, combining advanced fee markets, networking, and scaling architectures to support high-volume applications while preserving settlement guarantees.

- Delivers a next-generation fee market design and simulation framework enabling predictable, priority-aware transaction inclusion under varying network conditions. (TRL 2 > TRL 4–5)
- Supports data availability framework implementation for Layer 2 scaling to enable efficient off-chain execution and settlement. (TRL 3–6)
- Delivers a Cardano-specific sharding feasibility and protocol design study defining horizontal scaling pathways. (TRL 1 > TRL 2–3)
- Develops Byzantine-resilient networking models and a pub/sub prototype infrastructure with formal guarantees and implementation artefacts. (TRL 2 > TRL 4–5)
- Delivers L2 and ZK execution infrastructure including succinct proof architecture, L1 verification prototype, and zk-rollup security analysis. (TRL 2 > TRL 4–5)

2.1 Congestion Control & Transaction Efficiency

Cardano’s current transaction model provides predictable fees but lacks mechanisms for expressing urgency, optimizing resource allocation, and supporting emerging Layer 2 (L2) architectures. While the network is not currently experiencing sustained congestion, growth in DeFi activity and the emergence of rollups introduce new requirements around transaction prioritisation, data availability, and user experience. This stream therefore focuses on improving how network resources are allocated and consumed, ensuring both near-term usability and long-term scalability.

(i) On-Chain Data Availability for L2s

As Cardano's L2 ecosystem develops (eg, Hydra, Midgard), there is an increasing need for efficient on-chain data availability (DA) mechanisms. The work explores the design space for a DA layer that enables rollups and off-chain systems to post data to L1 at low cost, drawing lessons from approaches such as blob-based architectures. A central challenge is balancing cost efficiency with verification and storage requirements, ensuring that L2 scaling does not introduce new bottlenecks at the base layer.

(ii) Fee Market Design & Transaction Prioritisation

The current fee model does not provide explicit signalling of transaction urgency, which may become a limitation under future congestion scenarios. This work investigates tiered or dynamic pricing mechanisms within the Leios architecture, enabling differentiated inclusion guarantees while preserving fairness transaction inclusion. Given current network conditions and anticipated throughput increases, this work is exploratory and scoped, focusing on defining viable designs and simulations rather than full implementation.

2.2 Sharding

Cardano's current scaling roadmap – centred on Leios (throughput) and Hydra (off-chain execution) – addresses near- to medium-term demand. Sharding represents a longer-horizon exploration of horizontal L1 scaling through partitioning of ledger state. However, execution sharding remains an unresolved problem across blockchain systems, particularly due to challenges in cross-shard coordination and composability.

This work focuses on a single question: Is there a viable and differentiated approach to sharding that applies to the specifics of Cardano (EUTXO and Nakamoto-style consensus), *and is it worth pursuing beyond the current scaling roadmap?*

(i) Sharding Feasibility & Design Space Analysis

Existing sharding approaches will be evaluated, and sharding considered from first principles with a focus of optimizing it to the specifics of Cardano (EUTXO, etc.).

This workstream also aims at improving over the current state of art of sharding in general. A major issue, for instance, are cross-shard transactions that cannot be handled by one single shard, and thus cause protocol overhead. An objective will be to explore how the occurrence of cross-shard can be minimized.

2.3 Communication Infrastructure

Cardano's networking stack needs to evolve along two complementary directions: strengthening the formal guarantees of the existing peer-to-peer layer that carries consensus, and enabling a new class of authenticated application-layer messaging that the current stack does not serve.

This workstream addresses both a foundational hardening track on Ouroboros's network layer, and a new Cardano-anchored pub/sub protocol with a native operator-reward mechanism.

(i) Hardening the Ouroboros Network Stack

The first objective is to close known gaps in the Byzantine-resilience analysis of Cardano's peer-to-peer layer. Coretti et al. (CCS 2022) gave the first provably-secure gossip construction and showed that Ouroboros Praos remains secure over it. This track extends that analysis to realistic deployment conditions. Four axes are in scope: bandwidth constraints, with direct bearing on the security of Ouroboros under bounded capacity and on downstream high-throughput protocols such as Leios; inclusion of low-stake SPOs, relays, and client nodes as first-class participants without loss of Sybil resistance; formal treatment of eclipse-resistance; and a formalization of key components of the deployed network stack, bridging the theoretical model to the real-world implementation.

(ii) Pub/Sub Messaging System

The second track delivers a new Cardano-anchored publish/subscribe layer with best-effort security grounded in industry best practices. Identifiable publishers deliver messages and verifiable events to subscribers, with channels and access restrictions committed on-chain and Sybil resistance drawing on main-chain stake. Target scenarios of this authenticated notification primitive are governance notifications under CIP-1694, SPO-to-delegator communications, authenticated emergency alerts to SPOs, and ecosystem announcements. A paired effort focuses on the incentive layer that compensates operators for provable work (eg., forwarding and delivery) – designing a sound proof-of-participation scheme is itself a principal research question.

2.4 Layer 2 & ZK Verification Infrastructure (Succinct Proofs)

Cardano's Layer 2 ecosystem (eg, Hydra, Midgard, Gummiworm) is evolving toward higher throughput and specialized execution environments, but currently lacks a general-purpose mechanism for verifiable settlement on L1. At the same time, verifying non-trivial zero-knowledge (ZK) proofs on Cardano L1 remains impractical within existing execution budgets, limiting adoption across both rollups and general-purpose applications.

This program of work addresses these constraints by developing a unified ZK verification and proof infrastructure layer, enabling efficient on-chain verification of succinct proofs for both L2 settlement and broader application use. Additionally, a zk-Rollup (Hydra Tail) protocol is developed that makes use of this ZK infrastructure.

(i) Succinct Proof Systems for L2 Settlement

The first requirement is to define a proof system capable of compressing many off-chain state transitions into a small, verifiable proof. The work explores recursive SNARK constructions, including folding schemes and incremental verifiable computation, to enable scalable aggregation. The objective is to establish an architecture suitable for rollup-style L2

settlement on Cardano, rather than adopting a single proving system.

(ii) ZK Verification Infrastructure on Cardano L1

A primary barrier to the adoption of zero-knowledge (ZK) applications on Cardano is the development complexity, limited support to existing ZK building blocks, and cost of on-chain proof verification within the Plutus model, where proofs often exceed transaction limits or must be split across multiple transactions. This workstream establishes a practical ZK verification infrastructure on Cardano L1, enabling new proofs to be verified and unlocking use cases across L2 settlement, proof-of-computation, and identity. It focuses on identifying and standardizing previously unsupported SNARK primitives (eg., STARK-based), or adding expressiveness or efficiency to existing ones (Halo2/Plonk), alongside prototyping and benchmarking to ensure efficient execution within existing constraints.

In parallel, the work explores ecosystem interoperability challenges, particularly the mismatch between BN254-based ZK tooling and Cardano's BLS12-381 support, to unlock access to mature developer tooling and cross-chain applications. By translating these capabilities into developer-friendly tools and, where appropriate, CIPs, this stream lowers the barrier to ZK adoption, reduces verification costs, and enables a new class of scalable and privacy-preserving DApps. Impact will be measured by increased transaction volume leveraging ZK primitives and the number of ZK use cases executable within a single transaction.

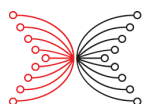
(iii) zk-Rollup Protocol and Analysis

Although zk-rollup protocols are widely used, their security is typically based on informal arguments rather than rigorous cryptographic proofs. This work addresses that gap by designing a provably secure zk-rollup protocol for UTxO-based ledgers, formalised within the Universal Composability (UC) framework. The goal is to define an ideal functionality and develop a protocol that realises it while balancing security, simplicity, and minimal mainchain footprint.

Out of scope: This scope excludes work proposed in the IO & Midgard Labs: L2 Scalability Initiative to implement a data-availability solution to satisfy immediate infrastructure needs.

WP3: Developer Platform & User Experience

Leads	Manuel Chakravarty, Marc Roeschlin, Carmine Ventre
FTE	5.7 (3 IOR, 2.7 ARC)
KPIs	3.1 Plutus: Adoption, Reliability <i>Monthly Active Users</i>: Improves developer-driven application usage through safer, more reliable smart contracts



	• <i>Monthly Uptime</i>: Reduces risk of contract-level failures through formally verified execution and compilation 3.2 Light Clients & Trustless APIs: Adoption • <i>Monthly Active Users</i>: Enables broader wallet and mobile adoption through reduced infrastructure requirements • <i>Monthly Transactions</i>: Increases transaction volume via faster onboarding and improved UX 3.3 Babel Fees Infrastructure: Adoption, Revenue • <i>Monthly Transactions</i>: Enables new transaction flows via fee abstraction and intent-based execution • <i>Annual Protocol Revenue</i>: Increases fee generation through higher-value and more accessible transactions
Pillars	P2: Adoption & Utility (P1: Infrastructure & Research Excellence) 3.1 Plutus • <i>I.2 Security & Resilience, Client diversity</i>: Enabling high-assurance smart contract development through formal methods and verifiable execution • <i>A.3 Developer experience, Open-source incentives</i>: Reducing friction in contract development and improving reliability for production applications 3.2 Light Clients & Trustless APIs • <i>I.1 Scalability & Interoperability, L2 integration</i>: Enabling lightweight interaction with Cardano without full node requirements • <i>A.2 Experience, Invisible technology</i>: Reducing reliance on centralised infrastructure and improving accessibility across devices 3.3 Babel Fees Infrastructure • <i>I.1 Scalability & Interoperability, L1 protocol improvements</i>: Supporting new transaction construction models and multi-asset fee mechanisms • <i>A.2 Experience, Invisible technology</i>: Reducing onboarding friction through fee abstraction and intent-based interactions

WP3 Developer Platform & User Experience enhances developer productivity and user accessibility through formalised tooling, abstraction layers, and lightweight access models, reducing friction and enabling scalable growth in applications and active users.

- Delivers a formally verifiable smart contract stack including state-machine development environments and verified compiler research to improve developer assurance. (TRL 2 > TRL 3–4)
- Enables next-generation light client infrastructure including intent-based transaction models and decentralized API/indexer prototypes. (TRL 2 > TRL 3–4)
- Produces Babel fee and intent execution market designs enabling multi-asset, abstraction-driven user interaction. (TRL 1 > TRL 3–4)
- Delivers minUTXO redesign proposals and CIP-ready recommendations to improve developer experience and state efficiency. (TRL 2 > TRL 4–5)

3.1 Plutus

Cardano's smart contract platform offers strong security guarantees, but developers face significant friction in building and deploying applications. These issues increase development time, introduce avoidable errors, and limit ecosystem growth. The objective is to provide a practical, verifiable foundation for developing safety-critical contracts, reducing reliance on ad hoc implementations, and enabling higher confidence in correctness for complex on-chain applications.

(i) Verifiable State Machine Models for Smart Contracts

Safety-critical smart contracts require formal mathematical reasoning to achieve high confidence in their correctness. This work builds on prior research demonstrating that state machine models provide an effective abstraction for specifying contract behaviour at a level suitable for formal verification.

The approach formalises these models in proof assistants such as Agda or Lean, enabling both rigorous reasoning about correctness properties and executable specifications via code extraction. This supports a development workflow in which contracts can be specified, tested, and validated within a single formal framework, facilitating rapid prototyping while maintaining strong assurance guarantees.

(ii) Verifying Optimizing Compiler

Research is supporting the engineering team to develop a verifying and optimizing compiler for Plutus. The verifying compiler issues a certificate for each optimization phase, which can be checked in Agda to ensure there exists a proof that the source and target of the optimizing phase are equivalent.

Scope clarification: IO Engineering will lead the delivery of the verifying and optimising compiler, with IO Research supporting its design and validation. In parallel, IOR will lead the publication of a peer-reviewed paper on the compiler, with engineering contributing—both outputs delivered as joint efforts across research and engineering.

3.2 Light Clients & Trustless APIs

Today, most Cardano wallets depend on centralized API providers to interact with the blockchain, introducing both centralization risk and operational fragility. While recent developments such as Mithril address chain synchronisation and state verification, a complete light client stack – covering transaction creation, data access, and interaction – does not yet exist.

This work addresses that gap by designing a trust-minimised light client architecture, enabling wallets and DApps to interact with Cardano efficiently without requiring full node infrastructure.

(i) Intent-Based Transaction Construction (Cavefish Protocol)

A central challenge in UTxO-based systems is transaction construction, which requires access to ledger state for UTxO selection, fee calculation, and change handling. Cavefish addresses this by introducing an intent-based protocol in which a light client expresses a desired action (eg, “send 10 ada”), and a service provider constructs a transaction that is verified and signed client-side. The protocol incorporates cryptographic primitives such as Blind Schnorr signatures and zero-knowledge proofs to ensure correctness, unforgeability, and privacy guarantees.

(ii) Decentralized Query Layer (Indexer/API Replacement)

To remove reliance on any single API provider, the stream designs a decentralized indexer and query layer that enables light clients to access historical data and UTxO information. This includes exploring incentive and fee mechanisms to ensure data availability and correctness, as well as defining trust, security, and performance trade-offs in the resource-constrained context of light clients.

(iii) Light Client Architecture & Focus

The work designs a coherent light client architecture which leverages transaction construction (Cavefish), data access (Indexer/API), and synchronisation (via Mithril). A central focus is ensuring fast time-to-first-interaction, low-latency event delivery, and compatibility with wallet use cases and requirements. Security proofs and formal methods are used where appropriate to validate correctness and security properties.

3.3 Babel Fees Infrastructure

Cardano’s requirement that transaction fees be paid in ada introduces friction for new users and enterprise participants, many of whom operate primarily in stablecoins or application-specific tokens. An engineering workstream is already delivering an initial Babel Fees infrastructure piece via CIP-118 and using a centralised clearing model to achieve near-term production readiness. The broader challenge, however, remains unresolved: how to design a scalable, decentralized fee market that operates effectively within Cardano’s EUTXO-based ledger.

This research focuses on the economic and protocol foundations required to evolve from this initial centralised model to a permissionless, multi-asset fee marketplace. It also addresses related usability and efficiency constraints, particularly the minUTXO requirement, which complicates token-only interactions. More broadly, the work supports Cardano’s transition toward intent-based interaction models, where users specify desired outcomes and third parties compete to execute them. Delivering this requires robust market design, aligned incentives, and resilience to manipulation.

(i) Decentralized fee-provider market design

The first objective is to define how a competitive, permissionless network of fee providers should operate. This includes modelling price formation, provider incentives, and mechanisms for matching user transactions or intents with agents willing to supply ada for fees in exchange for other assets. The central question is how to ensure efficiency, manipulation resistance, and alignment with Cardano’s decentralization goals.

(ii) Cardano Intent Solver Markets

Beyond simple fee abstraction, Babel Fees open a path toward broader delegated execution on Cardano. Instead of manually constructing fully specified transactions, users could express desired outcomes, while competing solvers determine how best to execute them through batching, routing, timing, and other execution strategies compatible with the EUTXO model. This workstream begins to study the market design, validation rules, and execution guarantees needed for such solver markets to function safely and efficiently on Cardano. Dimensions of interest include how users constrain solver discretion, how fulfilment is verified, and how harmful extractive behaviour can be limited.

(iii) minUTXO alternatives and UTxO efficiency

The minUTXO requirement is effective in controlling UTxO set growth but introduces significant friction for developers and users, particularly in multi-asset contexts. This work evaluates alternative mechanisms that preserve economic safeguards while improving usability – such as revised cost models, dynamic constraints, or more efficient resource accounting. The goal is to reduce transaction friction and simplify wallet and DApp design without weakening ledger discipline.

Out of scope: This scope aligns to but excludes the work in the IO: Cardano Upgrades proposal for the development and delivery of Babel Fees, including its Nested Transactions (CIP-118) framework, wallet integrations, routing, and production MVP roadmap.

WP4: Applications, Adoption, & Liquidity

Leads	Pooya Farshim, Luca Debiassi
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FTE	3.6 (2 IOR, 1.6 ARC)
KPIs	4.1 Bridges, Atomic Swaps & Agent Interoperability: Adoption, Revenue • <i>Total Value Locked</i>: Enables large-scale cross-chain capital inflows (eg, BTC, external assets) • <i>Monthly Transactions</i>: Drives activity from bridging, swaps, and cross-chain execution • <i>Annual Protocol Revenue</i>: Increases fee generation from high-value transactions and liquidity flows
Pillars	P1: Infrastructure & Research Excellence (P2: Adoption & Utility) 4.1 Bridges, Atomic Swaps & Agent Interoperability • <i>I.1 Scalability & Interoperability, Cross-chain interoperability</i>: Secure bridges, state proofs, and cross-chain execution primitives connecting Cardano to external ecosystems • <i>A.1 High-Value Verticals, DeFi</i>: Enabling institutional-grade liquidity inflows and cross-chain asset utilisation • <i>A.2 Experience, Enterprise security & compliance</i>: Providing verifiable, high-assurance cross-chain coordination and custody mechanisms

WP4 Applications, Adoption & Liquidity accelerates capital inflows by enabling secure interoperability, composable applications, and new interaction paradigms, positioning Cardano as a hub for cross-chain liquidity and enterprise use cases.

- Delivers formal bridge security models and specifications enabling trust-minimised cross-chain connectivity. (TRL 1 > TRL 2–3)
- Produces bridge verification benchmarks and prototypes demonstrating feasibility of Bitcoin and external integrations. (TRL 2 > TRL 4–5)
- Delivers n-party atomic swap prototype and formal model enabling composable, trustless liquidity flows. (TRL 1 > TRL 2–3)
- Delivers agent chain specification and prototype enabling Cardano to act as an autonomous cross-system coordination layer. (TRL 1 > TRL 2–3)

4.1 Bridges, Atomic Swaps, & Agent Chains

Bridges are the primary pathway for capital to enter the Cardano ecosystem, yet they remain one of the highest-risk components in blockchain systems. While multiple interoperability solutions are emerging – such as Bitcoin bridges, cross-chain messaging layers, and native atomic swaps – their security assumptions, adversarial resilience, and operational guarantees are not yet fully formalised or independently validated at a system level.

At the same time, blockchains remain largely passive systems, unable to act directly within external environments. Enabling Cardano to function as an active coordination layer – through secure cross-chain execution, threshold signing, and asset control – requires new cryptographic primitives and stronger formal guarantees.

This system-level work therefore focuses on delivering high-assurance security foundations and trust-minimal interoperability protocols that strengthen existing bridge infrastructure, enable trust-minimised cross-chain execution, and position Cardano as a reliable coordination layer across ecosystems. It also considers analyzing post-quantum threshold signatures, their application to cross-chain coordination and external system interaction.

(i) Formal Security Analysis of Bridge Infrastructure

This stream provides high-assurance, formal verification, and security analysis of the diverse bridge systems being deployed on Cardano. It defines explicit trust models (eg, threshold honesty assumptions), analyses adversarial behaviour and failure modes, and verifies key properties such as ownership preservation, atomicity, and liveness across chains.

Particular emphasis is placed on Mithril-based attestation, including the feasibility, cost, and security implications of verifying certificates in external environments, which is essential to Cardano's DeFi solutions. The objective is to produce auditable, formally grounded guarantees that can be relied upon by bridge operators, auditors, and institutional participants.

(ii) Atomic Swaps & Trust-Minimal Cross-Chain Execution Protocols

This stream develops trust-minimised mechanisms for cross-chain asset exchange without intermediaries. Using adaptor signatures and multi-party constructions, it enables atomic settlement across assets and chains, including 2-party and n-party swap protocols (eg, CANS).

These primitives form the foundation for intent-based cross-chain execution, where users express high-level outcomes, and solvers coordinate fulfillment through atomic transaction bundles. The work evaluates scalability, composability, and deployment feasibility, with the goal of providing secure, reusable building blocks for interoperability.

(iii) Agent-Based Cross-Chain Control & Coordination

This stream explores protocols that enable Cardano to act as a distributed cryptographic agent, where participants (eg, SPOs) jointly control keys and perform verifiable actions on external systems. This includes threshold signing, cross-chain transaction authorisation, and coordinated custody of external assets (eg, Bitcoin).

The work formally characterises trust assumptions, decentralization properties, and operational guarantees, and distinguishes this model from existing approaches such as Mithril (attestation), threshold custody systems (eg, tBTC), and chain-level signing paradigms (eg, NEAR Chain Signatures).

Validation is grounded in concrete use cases – such as Bitcoin transaction signing or cross-chain message authorisation – to assess feasibility and integration complexity. The objective is to determine whether this model provides a meaningful security or architectural advantage within Cardano’s interoperability stack.

Out of scope: This system-level scope excludes development work proposed in Cardano treasury bridge-related proposals including *Withdraw ₦700,000 for ZK Bridge*, administered by Intersect, and *Pogun: Capital Without Compromise*.

WP5: Economic Systems & Incentives

Leads	Matthias Fitz, Evangelos Markakis
FTE	3.7 (2.5 IOR, 1.2 ARC)
KPIs	5.1 SPO Incentives & Decentralization: Reliability, Resilience • <i>Monthly Uptime</i>: Aligns incentives to maintain consistent block production and network reliability • <i>Operational Resilience</i>: Improves decentralization and reduces concentration risk 5.2 Resource Diversification: Revenue, Resilience • <i>Annual Protocol Revenue</i>: Explores new revenue sources (multi-resource participation, shared security) • <i>Operational Resilience</i>: Diversifies resources securing the network beyond stake 5.3 Metrics: Adoption • <i>Monthly Active Users</i>: Enables better understanding of real user engagement
Pillars	P5: Ecosystem Sustainability & Resilience (P1: Infrastructure & Research Excellence, P3: Governance) 5.1 SPO Incentives & Decentralization • <i>E.2 SPO Incentives, Diversified SPO roles</i>: Incentivising participation across block production and emerging roles (eg, Mithril, Leios) • <i>E.2 SPO Incentives, Decentralization target</i>: Maintaining a healthy distribution of independent operators 5.2 Resource Diversification • <i>E.1 Financial Stewardship & Tokenomics, L2 > L1 value retention</i>: Exploring multi-resource participation and shared security models

	• <i>E.2 SPO Incentives – Diversified SPO roles</i>: Extending validator roles beyond block production 5.3 Metrics (Effective Throughput) • <i>I.1 Scalability & Interoperability, L1 protocol improvements</i>: Defining meaningful performance metrics beyond TPS • <i>P3 Governance</i>: Providing objective metrics for evaluating protocol changes
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WP5 Economic Systems & Incentives designs and validates sustainable, system-level economic models that align incentives across participants, ensuring decentralization, efficient capital utilisation, and long-term network viability.

- Delivers SPO incentive models and parameter recommendations to improve decentralization and participation efficiency. (TRL 4–6)
- Produces multi-resource consensus and restaking economic models enabling diversified security and capital efficiency. (TRL 1 > TRL 1–2)
- Delivers Proof-of-Useful-Work protocol design and analysis enabling productive resource utilisation in consensus. (TRL 1 > TRL 3–4)
- Develops new performance and utility metrics frameworks (eg, utility-weighted throughput) to support evidence-based optimisation. (TRL 1 > TRL 3–4)

5.1 SPO Incentives & Decentralization

Cardano’s stake pool incentive mechanism was designed for a simpler operating model in which SPOs were primarily responsible for block production. The network has since evolved: SPOs are now expected to support additional infrastructure and services (eg, Mithril signing and forthcoming Leios and Peras responsibilities), each introducing additional compute, bandwidth, and operational costs without corresponding adjustments to rewards.

(i) **Recalibration of SPO Incentives for Multi-Task Participation**

This stream focuses on recalibrating the incentive model to reflect the expanded role of SPOs and to ensure long-term decentralization and network resilience. The work models SPO cost structures under current and future workloads, defines the desired structure of the SPO ecosystem (eg, pool viability, distribution, and operator diversity), and evaluates how existing parameters (eg, k , pledge influence, reward curves) shape these outcomes.

5.2 Resource Diversification

Standard permissionless consensus models rely on a single primary resource – eg., ada stake – to provide security. In contrast, by using multiple consensus resources (eg., proofs of

useful work (PoUW) or restaking/actively validated services (AVS)), a system can maintain integrity even if one resource is compromised. This is especially useful for bootstrapping new blockchains initially facing low participation (low liquidity or uneven stake distribution, etc.). Leveraging existing assets – such as restaked ada – enables such early-stage networks to draw on stable participants for secure operation. Offering this capability on Cardano thus promises to attract sidechain (launches) to the Cardano ecosystem.

PoUW, as a special case for such an additional resource, executes real-world computational tasks as an additional effect besides securing consensus – in contrast to plain PoW, which serves no second purpose. Computing succinct state proofs as the useful work in PoUW is of special interest since they are required to support light clients and bridges, i.e., PoUW can be applied where the useful work directly benefits the underlying system, and the involved computational efforts are directly incentivized via block rewards.

(i) Multi-Resource Consensus

Maintains a focused research thread on multi-resource BFT consensus for Cardano sidechains, examining secure multi-resource re-staking equipped with robust recovery mechanisms for safety violations (due to corruption exceeding the BFT security threshold) beyond previously known guarantees, and guaranteeing optimal economic safety (implying the reliable identification and penalization of misbehaving parties). Furthermore, the incentives of multi-resource restaking and optimal economic safety are explored.

(ii) PoUW to Generate Succinct State Proofs for Cardano

Current solutions for PoUW do not allow for a parameterization that makes it fit for blockchain mining. This stream aims at closing this gap with the goal of obtaining a research paper describing a provably secure PoUW-mining protocol that can be used for Cardano (as an additional resource to stake) and its sidechains that generates succinct state proofs to support light clients and secure cross-chain interoperability.

5.3 Metrics

Traditional throughput metrics such as transactions per second (TPS) provide an incomplete and often misleading view of system performance, as they do not account for the varying computational complexity, data footprint, and economic utility of different transactions. As a result, TPS alone is insufficient for comparing ecosystems or evaluating the impact of protocol changes.

(i) Effective TPS Metrics

This stream develops a more meaningful notion of throughput by defining metrics that account for the utility delivered by transactions, including factors such as computational complexity, state changes, data storage, and system-level guarantees (eg, decentralization). The objective is to move beyond raw transaction counts toward a measure that reflects the actual value provided by the network as a decentralized computation and settlement system.

The work defines and compares alternative throughput measures, and applies them to evaluate Cardano and other major ecosystems (eg, Bitcoin, Ethereum) under consistent assumptions. The resulting framework enables like-for-like comparison across systems and across protocol configurations, and provides a basis for assessing the impact of changes such as Leios and Peras, as well as a future post-quantum version of Ouroboros.

Out of scope: This scope excludes work proposed in the IO & Ensurable Systems: Cardano Maintenance Initiative including mainnet and global mempool monitoring, fee estimation, ledger performance analysis, node optimisation, benchmarking for releases and hard forks, and distributed cluster operations.

WP6: Governance, Identity, & Social Infrastructure

Leads	Paolo Penna, Anca Nitulescu
FTE	2.5 (2.5 IOR)
KPIs	6.1 Governance: Governance, Resilience • <i>DRep Participation Rate</i>: Increases active participation and decision quality in governance • <i>Operational Resilience</i>: Reduces concentration risk and mitigates governance capture 6.2 Decentralized Identity: Adoption, TVL & Governance • <i>Monthly Active Users</i>: Enables identity-driven user engagement (eg, wallets, credentials, access control) • <i>Total Value Locked</i>: Supports institutional participation through compliance and identity primitives
Pillars	P3: Governance (P2: Adoption & Utility) 6.1 Governance • <i>G.1 Incentivized & Accessible Governance, Role-based incentives for DReps, SPOs & CC</i>: Improving participation incentives, accessibility, and engagement of DReps and stakeholders 6.2 Decentralized Identity • <i>A.2 Experience, Decentralized Identity (SSI)</i>: Enabling verifiable credentials, selective disclosure, and identity-enabled transactions

WP6 Governance, Identity & Social Infrastructure builds robust coordination and trust

frameworks, combining decentralized governance, identity systems, and participation mechanisms to enable scalable, secure, and inclusive ecosystem decision-making.

- Delivers governance incentive models and CIP-ready mechanisms to improve participation and decision quality. (TRL 2 > TRL 3–4)
- Develops governance DSL and prototype system enabling verifiable, programmable governance workflows. (TRL 1 > TRL 3–4)
- Delivers proof-of-personhood system and prototype enabling identity-linked governance and participation. (TRL 1 > TRL 3–4)
- Produces identity primitives and integration artefacts with CIP pathways supporting selective disclosure and compliance. (TRL 2 > TRL 4–5)

6.1 Governance

Cardano's on-chain governance system (CIP-1694 / Voltaire) is among the most advanced in the blockchain ecosystem, but current data indicates declining participation and increasing concentration of voting power. This creates a structural risk: if governance participation remains low and influence becomes concentrated, the system's ability to allocate treasury resources effectively and maintain legitimacy is weakened.

(i) Governance Incentives, Participation, & Verifiable Governance Logic

This stream addresses governance at two complementary layers. First, it designs incentive mechanisms and participation models to increase voter engagement, reduce concentration risk, and preserve decentralization while maintaining safety and liveness guarantees. Second, it improves how governance rules are specified and implemented by introducing a domain-specific language (DSL) for DAO governance, enabling governance logic to be expressed, audited, and verified more reliably.

The work evaluates alternative voting mechanisms (eg, delegation incentives, participation rewards), models their impact using on-chain data and agent-based simulations, and identifies parameter changes that can be implemented within the CIP-1694 framework. In parallel, it develops a constrained, formally grounded DSL for governance, aligned with existing models (eg, Agora), which compiles to the Aiken/Plutus stack (enhancing the Plutus program in WP4) and enables verifiable, auditable governance specifications.

6.2 Decentralized Identity

Cardano currently lacks a native mechanism to establish high-trust, Sybil-resistant identity on-chain. Without proof of personhood or verifiable credentials, governance remains stake-weighted, reputation systems cannot emerge, and compliance-sensitive applications are difficult to support.

This stream focuses on delivering privacy-preserving identity primitives through two priority

use cases, each grounded in deployable functionality and existing ecosystem infrastructure.

(i) Sybil-Resistant Governance (Proof of Personhood)

This track enables governance models that go beyond stake-weighted voting by introducing proof of personhood into the CIP-1694 framework. The objective is to support mechanisms such as one-person-one-vote or hybrid models, improving participation and reducing concentration risk while preserving decentralization and privacy. The work defines how personhood proofs integrate with existing governance components (eg, DReps, delegation, voting flows), and evaluates trade-offs between Sybil resistance, accessibility, and user privacy. Outputs are validated through a working prototype demonstrating identity-enabled governance.

(ii) Selective Disclosure for Compliance (Verifiable Credentials)

This track enables users to prove eligibility attributes, such as KYC status, residency, or accreditation, without revealing underlying personal data. The objective is to support compliance-sensitive applications while maintaining self-sovereign identity principles. The work develops verifiable credential flows using privacy-preserving techniques (eg, zero-knowledge proofs), and defines how these integrate into wallet and DApp interactions. Outputs are validated through a prototype demonstrating selective disclosure in a real application context.

WP7: Delivery, Dissemination & Partnerships

Leads	Fergie Miller
FTE	1.75 (1.25 IOR 0.5 ARC) in-kind / absorbed by IOG
KPIs	7.2 Dissemination, Communication & Exploitation: Adoption <i>Monthly Active Users</i>: Increases developer and user engagement through improved communication, onboarding, and tooling <i>Monthly Transactions</i>: Drives adoption of new capabilities through integration, awareness, and deployment pathways
Pillars	P2: Adoption & Utility (P4 Ecosystem Growth, P3 Governance, P1 Infrastructure & Research Excellence) 7.2 Dissemination, Communication & Exploitation <i>Developer & user enablement</i>: Translating technical outputs into accessible artefacts (explainers, docs, integrations) <i>Community & ecosystem engagement</i>: Driving awareness and uptake through R&D Sessions, content, and stakeholder engagement

	• <i>Commercialisation pathways:</i> Converting outputs (eg, CPS) into deployable artefacts (CIPs, integrations, products)
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WP7 Delivery, Dissemination & Partnerships establishes the governance, delivery, and communication backbone of the programme, ensuring coordinated execution across workstreams while translating research outputs into ecosystem adoption, integration, and deployment.

- Delivers programme governance, coordination, and end-to-end oversight, ensuring alignment with scope, timelines, budget, and dependencies.
- Provides consolidated reporting, milestone tracking, and performance monitoring to support transparency and informed decision-making.
- Delivers structured dissemination across blogs, explainers, digital channels, and R&D Sessions, ensuring outputs are accessible and understood by key stakeholders.
- Drives ecosystem engagement, partner integration, and exploitation by enabling interoperability, supporting onboarding, and progressing CPS artefacts into implementation-ready CIPs.

7.1 Project management

Project management and coordination establish the governance, delivery, and reporting structures required to ensure effective execution of the program across all partners and workstreams. This includes maintaining end-to-end oversight of delivery progress, managing dependencies across technical teams, and ensuring alignment with scope, timelines, and budget constraints.

The delivery function is responsible for structured risk management, issue escalation, and change control across the program. This includes risk management, consolidated reporting, milestone tracking, and ensuring outputs meet agreed quality and compliance standards. The objective is to provide a stable delivery backbone that enables all technical workstreams to operate effectively and remain aligned with program goals.

7.2 Dissemination, Communication, & Exploitation

This work translates research and technical innovation into ecosystem adoption and market impact. It ensures outputs from IOR, ARC, and CBU teams are clearly communicated, understood by stakeholders, and developed into actionable pathways for integration and commercialisation.

The scope combines ecosystem communication, partner integration, and deployment enablement. This includes structured dissemination through product explainers, blogs, and digital channels, alongside continued delivery of Cardano R&D Sessions to bridge research

and builder communities. Engagement with SPOs, DReps, developers, and ecosystem participants supports awareness and uptake of new capabilities.

In parallel, the work enables ecosystem integration by supporting connectivity with critical infrastructure providers, partner chains, and cross-chain applications. It also defines institutional onboarding pathways, ensuring organisations can engage with Cardano in a manner aligned with their technical, operational, and regulatory requirements.

A core component is the definition of exploitation pathways, translating research outputs (eg, CPS) into implementation-ready artefacts (eg, CIPs), and supporting their progression toward deployment.

Risks, Dependencies, and Assumptions

The Cardano Vision 2026 program introduces a range of technical, operational, and strategic risks as it continues to advance novel protocol designs towards deployment.

Technical Risks

The primary challenge is system complexity and integration at scale. As Cardano evolves into a multi-layer architecture, upgrades to consensus, cryptography, and execution become interdependent and difficult to fully model under real-world conditions. At the same time, introducing new paradigms – such as post-quantum cryptography, zero-knowledge verification, and cross-chain execution – creates uncertainty across performance, security, and operational reliability.

Core risks fall into four groups:

- **Protocol & Cryptography Risk:** Uncertainty around MEV and transaction ordering, non-linear interactions between upgrades (eg, Leios, Peras), and risks in replacing core cryptographic primitives, including performance overheads from post-quantum designs and limits of on-chain ZK verification.
- **Interoperability & Execution Risk:** Dependence on correctly specified trust models for bridges and cross-chain systems, alongside coordination and reliability risks introduced by intent-based execution and solver architectures.
- **Economic & Incentive Risk:** Potential misalignment in SPO incentives and multi-resource models, creating unintended behaviours or centralisation pressures.
- **System & Operational Risk:** Increasing integration complexity across consensus, ZK, identity, and interoperability layers, combined with operational constraints such as hardware requirements (eg, HSMs), resource demands, and developer friction from the EUTXO model.

Additional risks arise from scaling trade-offs and adoption uncertainty. Data availability, sharding, and communication layers introduce cost and composability challenges, while new paradigms (eg, intents, identity, light clients) must balance decentralization with usability.

Simulation and modelling may not fully capture real-world or adversarial dynamics, and fragmentation across tooling ecosystems and emerging standards introduces longer-term uncertainty.

Key Dependencies & Assumptions

Dependencies are primarily related to alignment across the Cardano stack and ecosystem, including product roadmap coordination, integration with core infrastructure, and engagement with layer 2, SPOs, and DeFi partners.

The program assumes that adoption is driven by UX, liquidity, and infrastructure rather than throughput, that demand for L2s and interoperability will grow, and that new capabilities will be adopted where they are practical, integrated, and aligned with real use cases.

Strategic Risks & Mitigation

Strategic risks centre on misalignment with demand, adoption failure, and execution gaps between research and deployment. There is a risk of building ahead of clear need, particularly in advanced infrastructure, while increasing system sophistication may reduce usability and slow adoption. In parallel, failure to deliver high-assurance interoperability could limit liquidity inflows and institutional engagement – critical to Cardano’s 2030 objectives of increasing adoption and on-chain economic activity.

Mitigation is deliberately focused on execution discipline and demand alignment:

- Early product validation: Scope work against real use cases and partner demand from the outset
- Stage-gated delivery (TRL): Enforce clear progression criteria and stop/continue decisions
- Clear translation pathway: Structure outputs from research through to engineering delivery
- Build in public: Use R&D sessions, open prototypes, and explainers to validate direction and drive adoption

This approach shifts the program from exploratory research toward a controlled, integration-led delivery model, reducing execution risk while increasing the likelihood of measurable ecosystem impact.

Differentiation: Beyond State-of-the-Art

Cardano Vision 2026 advances beyond the state of the art by addressing system-level constraints across security, scalability, and usability, focusing on solutions that are both novel and deployable in real-world, decentralized environments.

Trust, Security, & Reliability Infrastructure

This work extends the frontier of protocol security and resilience by addressing challenges that remain unresolved in existing systems:

- *Enhanced consensus validation*: Evaluation of Leios and Peras under degraded and attack conditions, rather than optimistic assumptions
- *EUTXO-specific mitigations of the negative impact of MEV*: Structured analysis of MEV in deterministic UTXO systems, where ordering and contention differ fundamentally from account-based models
- *Post-quantum Ouroboros*: Credible pathway and constructions to quantum-secure PoS, including the major challenge of practical post-quantum VRFs
- *Production-grade key security*: Integration of HSM and MPC-based key management into live node operation

Scalability & Execution

Scalability is treated as a system-level constraint across execution, networking, and verification, providing the following impact:

- *EUTXO-native scaling architecture*: Data availability and execution models tailored to UTXO systems rather than adapted from account-based designs
- *Practical ZK verification on L1*: Focus on feasibility within real execution limits, enabling deployable ZK-based systems
- *Network-layer formal security*: Extension of protocol guarantees to realistic P2P conditions (eg, bandwidth constraints, eclipse resistance)

Developer Platform & User Experience

This program establishes a verifiable foundation for smart contract development through formal specifications and execution models, ensuring high-assurance contract design. This is reinforced by a verifying compiler pipeline, introducing proof-carrying optimisation and moving beyond best-effort correctness. At the interaction layer, the program establishes a robust babel-fees marketplace and focuses on intent-based execution models, shifting user interaction from transaction construction to outcome-based behaviour with little friction.

Applications, Adoption & Liquidity

The focus of this program is to enable trust-minimised cross-chain execution through atomic swaps and coordination primitives that remove reliance on intermediaries. This is complemented by agent-based interoperability models, allowing Cardano to operate across external systems through distributed key control.

Economic Systems & Incentives

This program introduces multi-resource consensus, extending security beyond stake and

enabling new forms of economic participation aligned with evolving network roles.

Governance, Identity & Social Infrastructure

This program delivers privacy-preserving identity primitives based on zero-knowledge credentials, enabling selective disclosure and supporting Sybil-resistant participation in governance and other sensitive applications. This provides the ability to share only what is necessary, while ensuring integrity in sensitive applications.

Activity of Work

WP1: Trust, Security & Reliability Infrastructure

1.1 Consensus

T1: MEV, Transaction Ordering & Fee Market Analysis	
T1.1 MEV Vector Identification and Modelling T1.2 Mempool Partitioning & Attack Analysis	D1.1 MEV & Mempool partitioning Assessment [Report]
T2: Protocol Validation & Operational Boundaries	
T2.1 Adversarial Modelling & Simulation T2.2 Performance & Resource Modelling T2.3 SPO Operational Boundaries T2.4 Formal Verification T2.5 Integration & Implementation Guidance	D2.1 Linear Leios Security & Performance Analysis [Report]
T3: Targeted Protocol Optimisation (Leios & Peras)	
T3.1 Peras Cooldown Avoidance T3.2 Voting Certificate Optimisation T3.3 Leios–Peras Interaction & Unification	D3.1 Protocol improvements on at least one of: (a) Peras Protocol Improvement for Avoiding Cooldowns (b) Voting Certificate Design and Analysis [Paper/Report] D3.2 P+L Unification Design and Analysis [Report]

1.2 Post Quantum

T1: Post-Quantum Cryptographic Primitives
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T1.1 Preliminary PQC Candidate Evaluation (signatures, VRFs, threshold schemes) for Cardano T1.2 Post-Quantum VRF Design & Analysis T1.3 VRF Performance Trade-off Assessment (proof size, verification cost, composability)	D1.1 Post-Quantum Cryptography Assessment of Primitives for Cardano [CPS + Report] D1.2 Post-Quantum VRF Design and Proof of Security [Paper]
T2: Protocol Integration & Validation	
T2.1 Quantum-Secure Ouroboros Modelling T2.2 Formal Security Analysis under Quantum Adversaries T2.3 Prototype Implementation of PQC Components within Ouroboros T2.4 Benchmarking & Performance Evaluation	D2.1 Quantum-Secure Ouroboros Security Analysis [Paper] D2.2 PQC Integration Specifications & Performance Benchmarks [Report]
T3: Migration Strategy & Ecosystem Readiness	
T3.1 Evaluation of Migration Strategies (full replacement vs incremental approaches) T3.2 Assessment of Compatibility & Ecosystem Impact (wallets, SPOs, addresses) T3.3 Transition Pathway Design (phasing, timelines, dependencies) T3.4 Roadmap Development & Positioning	D3.1 Cardano Post-Quantum Readiness Roadmap & Migration Strategy [Report + CIP]

1.3 Node Security

T1: Node Threat Model & Key Architecture Analysis	
T1.1 Identification and classification of node key types (KES, VRF, operational keys) and their roles T1.2 Threat modelling across software, hardware, and operational environments (eg, host compromise, memory exposure, key exfiltration) T1.3 Analysis of key lifecycle risks (generation, storage, rotation, usage) T1.4 Assessment of post-quantum implications for node-level key management T1.5 SPO consultation to validate operational constraints, usability requirements, and deployment realities	D1.1 Node Security Threat Model & Key Management Analysis [Report/Paper]
T2: HSM Integration for Secure Operational Key Management	

T2.1 Integration of HSM support into cardano-node for KES and VRF signing (eg, via PKCS#11)
T2.2 Extension of KES agent architecture to support hardware-backed key storage and signing

D2.1 HSM-Enabled Node Implementation (KES & VRF Signing) & Documentation [CPS + Report]

WP2: Scalability & Execution Layer

2.1 Congestion Control & Transaction Efficiency

T1: Fee Market Design & Transaction Prioritisation

T1.1 Analysis of transaction prioritisation needs under Leios
T1.2 Design of tiered or dynamic fee mechanisms
T1.3 Simulation of fee behaviour under realistic load scenarios
T1.4 Assessment of impact on fairness and predictability

D1.1 Fee Market Design Specification (Leios-Compatible) [Report]

D1.2 Simulation Results & Behavioural Analysis [CPS + Report]

T2: On-Chain Data Availability for L2s

T2.1 Requirements gathering with L2 builders
T2.2 Evaluation of DA design approaches (eg, blob-style architectures)
T2.3 Trade-off analysis (cost, storage, verification)
T2.4 Definition of Cardano-specific DA design space

D2.1 Data Availability Requirements & Design Framework [CPS + Report]

2.2 Sharding

T1: Sharding Feasibility & Design Space Analysis

T1.1 Evaluation of existing sharding approaches
T1.2 Develop a sharding protocol that is optimized for the specifics of Cardano
T1.3 Prove the protocol's security

D1.1 Evaluation of existing solutions and provably secure Cardano-optimized protocol [CPS + Report]

2.3 Communication Infrastructure

T1: Hardening the Ouroboros network stack

T1.1 Devise a bandwidth-constrained network model and corresponding diffusion strategies T1.2 Extend Ouroboros security analysis to the new model T1.3 First-class inclusion of low-stake SPOs, relays, and clients into the above model T1.4 Formal analysis of eclipse-resistance and formalization of other key components of deployed Cardano network stack	D1.1 Nakamoto blockchains in the face of Byzantine network adversaries and capacity constraints [Paper]
T2: Byzantine-resilient pub/sub	
T2.1 Proof-of-concept implementation with best-effort, best-practices security T2.2 Simulation and evaluation of above implementation T2.3 Develop incentive and proof-of-participation mechanism	D2.1 Pub/sub spec and prototype with best-effort security [CPS + Proto Docs]

2.4 Layer-2s & Succinct Proof Infrastructure (ZK-Enabled Settlement)

T1: Proof System Design & Architecture	
T1.1 Evaluation of recursive SNARK and folding scheme approaches T1.2 Design of proof aggregation model for L2 state transitions T1.3 Assessment of compatibility with Cardano execution constraints T1.4 Selection of candidate architecture for implementation T 1.5 Research and Analysis of a zkRollup protocol (“Hydra Tail”)	D1.1 Proof System Design & Trade-off Analysis [Report/Paper]
T2: ZK Verification Infrastructure & Primitive Integration	
T2.1 Identify and validate cryptographic primitives and interoperability constraints (eg, BN254 vs BLS12-381) T2.2 Design verification primitives, protocol extensions, and developer tooling for efficient single-transaction proof verification T2.3 Prototype and benchmark solutions to optimise performance within Plutus constraints	D2.1 ZK Verification Infrastructure for Cardano L1 [Proto Docs + Benchmark + CIP]
T3: zk-Rollup Protocol	

T3.1 Implementation of zk-Rollup protocol “Hydra Tail”	D2.1 zk-Rollup [Report/paper]
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WP3: Developer Platform & User Experience

3.1 Plutus

T1: Verifiable State Machine Model	
T2.1 Formalization of EasySM in Agda T2.2 Executable EasySM contract harness T2.3 Contract testing pipeline through code extraction to Haskell T2.4 Plutus contract generation from Agda specifications	D1.1 Research prototype of an EasySM development environment in Agda [Report/Paper + Proto Docs]
T2: Verifying Optimising Compiler	
T2.1 Formalize optimization correctness and design certificate generation/verification (eg, in Agda) T2.2 Integrate and validate the certification framework within the Plutus compilation pipeline	D2.1 Verifying Optimising Compiler Specification & Certification Framework [Report/Paper]

3.2 Light Clients & Trustless APIs

T1: Cavefish Protocol & Transaction Construction (IOR + ARC)	
T1.1 Formal specification of the Cavefish protocol T1.2 Implementation of intent-based transaction construction and verification T1.3 Integration of cryptographic primitives (WBPS, ZK circuits) T1.4 prototype (single service provider) T1.5 Benchmarking of transaction construction latency, correctness, and privacy guarantees	D1.1 Cavefish Protocol [Report/Paper] D1.2 Specification & Prototype [Proto Docs] D1.3 Network Prototype & Benchmark Results [Proto Docs]
T2: Decentralized Query Layer & Integrated Light Client Access	
T2.1 Design of decentralized indexer and query architecture T2.2 Evaluation of incentive and fee mechanisms T2.3 Implementation of prototype query layer	D2.1 Decentralized API / Indexer Specification, Prototype & Benchmarks [Report Paper + Proto Docs]

T2.4 Benchmarking of query performance and reliability under realistic workloads

3.3 Babel Fees Infrastructure

T1: Decentralized Babel Fees Marketplace Design

T1.1 Definition and comparison of fee-provider roles, market structures, and interaction models
T1.2 Modelling of quote formation, spreads, and price discovery across competing fee providers
T1.3 Analysis of provider incentives, concentration risks, and protections against manipulation, griefing, and extractive routing
T1.4 Simulation of marketplace behaviour under realistic network conditions
T1.5 Assessment of implementation pathways under current engineering roadmap (eg., nested transactions / CIP-118) and identification of any additional protocol or standards requirements

D1.1 Babel Fees Marketplace Design Specification and Recommended Market Mechanism [Report/Paper]

T2: Cardano Intent Solver Markets

T2.1 Specification of user outcome requests and solver discretion under EUTXO constraints
T2.2 Modelling of solver competition in Cardano, including batching, routing, timing, and inventory-based execution strategies
T2.3 Initial compatibility analysis with wallet UX, signing flows, and DApp interaction models

D2.1 (i) Desiderata for a Cardano Intent Solver Market (ii) Initial exploration of Wallet/DApp Integration [Report]

T3: minUTXO Alternatives & UTxO Efficiency

T3.1 Analysis of current minUTXO constraints, their economic roles, and sources of user and developer friction
T3.2 Exploration of alternative (pricing, accounting, or structural) mechanisms
T3.3 Evaluation of trade-offs (security, state growth, usability)
T3.4 Recommendation and refinement of the most promising approach(es), with implementation constraints

D3.1 minUTXO Problem Definition & Alternatives Report, Trade-off Analysis, and Policy Recommendations [CPS + Report + CIP]

WP4: Applications, Adoption, & Liquidity

4.1 Bridges, Atomic Swaps, & Agent Chains

T1: Formal Verification & Security Analysis of Bridge Infrastructure	
T1.1 Architecture Formalisation T1.2 Trust Model Definition & Assumption Mapping T1.3 Adversarial Modelling & Threat Analysis T1.4 Security Property Verification T1.5 Mithril Certificate Verification Benchmarking T1.6 Operator & Ecosystem Alignment	D1.1 (i) Mithril Bridge Security Model, Analysis, and Automated Model Checking [Paper/Report] (ii) Mithril Verification Benchmarks [Proto Docs] (iii) Bridge Security and Adversarial Threat Model Security Analysis [Report]
T2: Atomic Swaps Design and Prototype	
T2.1. State-of-the-art review T2.2. Protocol architecture design T2.3. Prototype development T2.4. Formal modelling & security analysis T2.5. Scientific and community dissemination	D2.1 (i) Prototype n-party swap case and security analysis [Report + Proto Docs]
T3: Agent Chain Specification & Prototype Validation	
T1.1 Formal definition of threshold signing model for external systems T1.2 Comparative analysis vs Mithril, tBTC, and NEAR Chain Signatures T1.3 Design of protocol architecture and trust model T1.4 Implementation of prototype for a concrete use case (eg, Bitcoin signing or cross-chain messaging) T1.5 Documentation of integration pathways for ecosystem partners	D3.1 Agent Chain Specification, Prototype & Integration Guide [CPS + Report/Paper + Proto Docs]

WP5: Economic Systems & Incentives

5.1 SPO Incentives & Decentralization

T1: SPO Incentive Model Assessment & Parameter Design

T1.1 Evaluate current SPO reward mechanism, parameters (eg, k, pledge, taskbased rewards) and their impact T1.2 Model SPO cost structures under current and future workloads (Mithril, Leios, Peras) T1.3 Define target SPO ecosystem characteristics (viability, distribution, decentralization) T1.4 Engage SPOs and stakeholders to validate assumptions and outcomes	D1.1 SPO Incentive Model Assessment & Parameter Recommendations [Report/Paper]
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5.2 Resource Diversification

T1: Multi-Resource Consensus Modelling	
T1.1 Multi-resource BFT consensus under weaker network assumptions, with improved recovery and economic safety T1.2 Incentive design for restaking and multi-resource participation T1.3 Extensions to Ouroboros for specialised SPO roles (eg, state proofs, archival services) T1.4 Proof-of-useful-work mechanisms for state-proof generation T1.5 Feasibility assessment and scale-up trigger conditions	D1.1 (i) Multi-resource BFT consensus [Report/Paper], feasibility assessment] (ii) Incentive mechanisms for restaking and multi-resource consensus [Report/Paper] (iii) Paper on provably secure PoUW for state-proof generation [Report/Paper+CPS]

5.3 Metrics

T1: Effective TPS Metrics	
T1.1 Definition of throughput metrics incorporating transaction utility (eg, computation, storage, composability) T1.2 Qualitative comparison of proposed measures against traditional TPS and existing alternatives T1.3 Design and implementation of measurement methodology and tooling T1.4 Quantitative evaluation of throughput across selected ecosystems (eg, Cardano, Bitcoin, Ethereum) T1.5 Application of metrics to Cardano under different configurations (eg, current state, Leios, Peras)	D1.1 Utility-Weighted Throughput Framework & Benchmarking Analysis [Report/Paper]

WP6: Governance, Identity & Social Infrastructure

6.1 Governance

T1: Governance Incentives & Participation Mechanisms	
T1.1 Analyse governance participation and voting power distribution (on-chain) T1.2 Design and evaluate alternative voting and incentive mechanisms T1.3 Model governance outcomes via simulation and agent-based approaches T1.4 Identify parameter changes within the CIP-1694 framework T1.5 Engage DReps, SPOs, and community to validate proposals	D1.1 Governance Incentive Models & Mechanism Assessment [Report/Paper + CIP]
T2: Domain-Specific Language for DAO Governance	
T2.1 Design constrained DSL for governance (Agora-aligned) T2.2 Develop formal semantics for verification of governance properties T2.3 Implement compilation pipeline to Aiken / Plutus with libraries T2.4 Integrate with ecosystem tooling (Mesh, Tx3, Blockfrost) T2.5 Deploy prototype governance flows (proposals, voting, treasury) T2.6 Validate with DAO builders and ecosystem partners	D2.1 DAO Governance DSL Specification, Prototype & Integration Toolkit [CPS + Paper + Proto Docs + CIP]

6.2 Decentralized Identity

T1: Sybil-Resistant Governance (Proof of Personhood Integration)	
T1.1 Definition of proof-of-personhood model aligned with Cardano governance requirements T1.2 Design of voting mechanisms enabling one-person-one-vote or hybrid models within CIP-1694 T1.3 Integration with existing governance infrastructure (DReps, delegation, voting flows)	D1.1 Proof of Personhood Model, ZK Identity Primitives & Governance Prototype [CPS + Paper + Proto Docs]

T1.4 Integration with Identus and privacy-preserving components (eg, Midnight where applicable) T1.5 Development of prototype demonstrating Sybil-resistant voting (eg, DRep voting with personhood constraint) T1.6 Evaluation of security, privacy, and participation trade-offs (eg, resistance to Sybil attacks vs accessibility)	
T2: Selective Disclosure for Compliance (Verifiable Credentials)	
T2.1 Define credential models (eg, KYC, residency, accreditation) T2.2 Design privacy-preserving proof flows (eg, ZK) T2.3 Integrate with Identus and wallet UX T2.4 Define compliance interaction flows (eg, eligibility checks) T2.5 Develop prototype for selective disclosure use cases T2.6 Assess regulatory alignment and interoperability	D1.2 Identity Primitives, Integration Specification [Paper + Proto Docs + CIP]

WP7: Delivery, Dissemination & Partnerships

7.1 Project Management

T1: Program Management and Coordination	
T1.1 Coordinate delivery across work packages, managing dependencies and timelines T1.3 Monitor progress, risks, and issues with structured mitigation and escalation T1.3 Manage resource allocation and budget tracking across partners T1.4 Lead stakeholder engagement, communications, and reporting (internal and external) T1.5 Ensure compliance with program requirements and quality standards	D1.1 Program governance and bi-annual reporting [Report]

7.2 Dissemination, Communication & Exploitation

T1: Dissemination & Ecosystem Engagement

T2.1 Deliver structured communication outputs (blogs, explainers, technical summaries, GitHub repos) T2.2 Deliver Cardano R&D Sessions and ecosystem-facing events T1.3 Coordinate with product, engineering, and innovation teams to enable delivery T1.4 Engage SPOs, DReps, builders, and ecosystem partners to drive awareness and adoption	D1.1 Program governance and bi-annual reporting [Report]
T2: Integration & Commercialisation Pathways	
T2.1 Translate research outputs (eg, CPS, papers) into implementation artefacts (eg, CIPs, specs) T2.2 Identify and prioritise integrations with key infrastructure providers T2.3 Support progression from prototype to deployment with engineering and product teams	D1.1 Program governance and bi-annual reporting [Report]

Key Metrics, Milestones, and Timeline

The CV26 program adopts a three-layer delivery model, progressing from foundational research through to ecosystem adoption. This reflects the IOR pipeline, foundational research → technology innovation → implementation readiness, ensuring outputs are systematically translated into deployable capabilities and measurable impact.

1. Novel Science (Papers and reports)

The program delivers a substantial body of novel science across ~38 papers and technical reports, providing the formal foundations required for protocol evolution. This includes 6 core publications, with 8 or more additional papers expected over the program lifecycle. Research is concentrated in high-impact areas, including:

- **Post-quantum cryptography:** Quantum-secure primitives (eg, VRFs, signatures) and migration pathways for Ouroboros
- **Communication infrastructure:** Byzantine-resilient network modelling under real-world constraints (eg, bandwidth, eclipse resistance)
- **Plutus & formal methods:** EasySM and formal execution models for verifiable smart contracts
- **Consensus & protocol design:** Adversarial analysis of Leios and Peras, including MEV and mempool dynamics
- **ZK & verifiable computation:** Succinct proofs, recursive systems, and on-chain verification constraints
- **Interoperability:** Formal security models for bridges, atomic swaps, and cross-chain coordination

- **Economic systems & governance:** Incentive design, multi-resource consensus, and privacy-preserving identity systems

2. Innovation Validation (CPS & Prototypes)

This layer focuses on demonstrating feasibility and bridging research to implementation. It includes ~9 Cardano Problem Statements (CPS), which define implementation pathways and requirements, and ~12 prototypes (approximately 25% of total outputs) providing technical validation. Key validation areas include:

- **Node security:** HSM-enabled node CPS and prototype integration for secure KES/VRF key management
- **ZK & Layer 2 infrastructure:** ZK verification toolkit and L2 settlement prototypes, enabling practical on-chain proof verification
- **Plutus & developer tooling:** EasySM prototype for verifiable smart contract development, alongside formal execution and testing frameworks
- **Light clients:** Cavefish protocol CPS, and prototype, including decentralized API/indexer solutions for trust-minimised wallet interaction
- **Interoperability:** Mithril bridge verification prototype, atomic swap prototype, and agent chain specification and prototype for cross-chain coordination
- **Identity:** Proof-of-personhood CPS and prototype, alongside verifiable credential and selective disclosure identity systems

Prototypes provide end-to-end and component-level validation across L2 settlement, ZK verification, light client stacks, and cross-chain systems. CPS outputs define the formal problem statements and implementation pathways. Together, these ensure outputs are technically feasible, benchmarked, and aligned with real system constraints.

3. Implementation-Ready (CIPs)

This layer represents the primary pathway to deployment. Around 5 Cardano Improvement Proposals (CIPs) formalize protocol changes and enable engineering integration, including three within the strategic focus of *Human Centred*, *Scalable*, *Post-Quantum Secure*:

- **Governance & identity:** Identity primitives, compliance use cases (eg, KYC), improved developer experience (EasySM), strengthened economic alignment (eg, SPO incentives), and reduced friction (eg, minUTXO)
- **L2 scaling & execution:** Consensus improvements, execution scaling (eg, L2s, data availability), ZK-enabled scaling (eg, rollups), and longer-term scaling research
- **Post-quantum readiness:** Long-term cryptographic resilience and migration pathways
- **minUTXO under Babel fees:** Efficiency improvements and policy recommendations
- **Governance incentives & DSLs:** Participation design and verifiable governance systems

Program Timeline & Prioritization

Given the exploratory nature of early-stage research, the program adopts a portfolio-based approach to resource allocation and prioritization. This includes a continuous assessment of progress, risk, and opportunity across workstreams, dynamic reallocation toward the highest-impact and most promising areas, and acceleration of high-performing workstreams and deprioritisation of those with limited alignment or impact.

This approach ensures that the program remains strategically focused, responsive to new insights, and aligned with ecosystem needs, while maintaining regular communication and formal review checkpoints. This is managed through a structured, transparent quarterly delivery updates and bi-annual reporting to the community via Intersect :

- **Cardano R&D session (webinar):** A public facing webinar outlining the WP26 scope, its rationale, strategic focus, initial progress on delivery and call for collaboration partners.
- **Mid-year review (interim report):** Provides a consolidated view of progress across workstreams, including task completion, deliverables, and minor change requests. Major changes are escalated and reviewed in coordination with governance bodies.
- **Q3 Cardano R&D session (webinar):** A public facing webinar outlining progress on each work package, planned outputs, innovation funnel maturation and implementation-readiness.
- **Year-end report (final report):** Delivers a comprehensive summary of outputs, including all deliverables, outcomes, and a full financial breakdown, ensuring accountability and transparency.

Together, these checkpoints ensure that progress is clearly communicated and that the program remains optimized for Cardano's strategic objectives.

2. Team

This proposal is submitted by a *9-partner consortium* led by Input Output Group (IOG), bringing together a network of leading academic institutions and industry partners with deep expertise across cryptography, distributed systems, formal methods, and blockchain engineering.

The consortium builds on the long-standing Blockchain Technology Laboratory (BTL) network and wider research collaborations, combining scientific excellence with applied engineering capability to deliver high-assurance research and implementation pathways for the Cardano ecosystem.

IOR: comprises 50+ researchers and architects across cryptography, distributed systems, formal methods, and crypto-economics, and collaborating with 20+ universities globally. Professor Aggelos Kiayias, Chief Scientist at IOG and Chair in Cyber Security and Privacy at the University of Edinburgh, is the architect of the Ouroboros protocol and a recognised

leader in blockchain research (FRSE, BCS Lovelace Medal 2024, ACM Fellow 2025, IACR Fellow, 2026). IOR operates through interdisciplinary teams combining research and applied engineering, including product managers, technical architects, applied cryptographers, and formal methods engineers, enabling a structured pathway from research to implementation.

University of Edinburgh: A leading global research institution and central node in the BTL network, with world-class expertise in cryptography, distributed systems, and formal methods. Key initiatives include the Edinburgh Decentralisation Index (EDI), and advanced research in zero-knowledge systems through the ZK lab.

Institute of Science Tokyo: A top-tier engineering institution (formerly Tokyo Tech), with strong research capabilities in blockchain, smart contracts, and system security. The institute contributes to protocol design, scalability, and secure distributed systems.

University of Buenos Aires: Argentina's leading university and globally ranked institution, with strong expertise in formal verification, distributed systems, and domain-specific languages. IOG has co-founded a distributed systems laboratory with the university to deepen collaboration within the Cardano ecosystem.

University of Sydney: A globally ranked research institution with strengths in cryptography, distributed systems, and blockchain through the Sydney Blockchain Group. Research spans secure Web3 infrastructure, DeFi systems, and AI-driven smart contract security.

University of California, Berkeley: Through its Department of Electrical Engineering and Computer Sciences (EECS), Berkeley is a global leader in cryptography and computer security. Research covers zero-knowledge proofs, privacy-preserving systems, and secure distributed computation, with significant impact across academia and industry.

University of Oxford: A leading centre for algorithmic game theory and computational economics. Research focuses on mechanism design, social choice, and incentive systems critical for governance, tokenomics, and decentralized coordination.

ZHAW Zurich University of Applied Sciences: Through its applied cryptography collaborations – (linked to the Zurich research ecosystem), ZHAW contributes expertise in provable security, post-quantum cryptography, and real-world protocol analysis, bridging theory and deployment.

Eryx: A worker-owned software cooperative with 15+ years of experience, specialising in zero-knowledge and blockchain technologies. Eryx contributes to advanced cryptographic tooling, including Aiken ZK, Proof of Innocence, ZK Login, and ZK Bridge, and brings practical experience in deploying SNARK/STARK-based systems.

Key Personnel

The IOR network combines world-leading expertise across cryptography, distributed systems, economics, and formal methods, alongside product, engineering, and commercial

leadership, ensuring delivery from research through to implementation and ecosystem adoption.

- **William Wolff, Lead Technical Architect (WP1).** Lead Technical Architect at Input Output, specialising in protocol design and high-throughput blockchain systems, leading multidisciplinary teams across research, simulation, and implementation.
- **Varun Maram, Research Fellow (WP1).** Assistant Professor at the University of Warwick and consultant at Input Output, specialising in post-quantum cryptography and provable security of real-world cryptographic systems, including contributions to NIST PQC candidates.
- **Matthias Fitzi, Principal Research Fellow (WP2, WP5).** Expert in fault-tolerant distributed protocols and provable security. PhD from ETH Zurich, with prior positions at UC Davis and Aarhus University. At Input Output since 2018, contributing to scalability initiatives including Leios and Hydra.
- **Sandro Coretti-Drayton, Senior Research Fellow (WP2).** Cryptographer specialising in scalable blockchains, consensus, and Byzantine-resilient networking. PhD from ETH Zurich, with postdoctoral research at NYU. Focuses on high-assurance distributed protocol design.
- **Jesús Díaz Vico, Applied Cryptography Lead (WP2).** Cryptographer specialising in blockchain, digital identity, and privacy-preserving systems. PhD with 20 years of R&D experience across industry and government. Former Research Engineer at IBM Research Zürich.
- **Carmine Ventre, Senior Research Fellow (WP3).** Professor at King's College London specialising in tokenomics, mechanism design, and computational finance, with a focus on economic systems for decentralized environments.
- **Pooya Farshim, Research Fellow (WP4).** Associate Professor of Cryptography specialising in provable security and cryptographic protocol design, with prior roles at ENS Paris and TU Darmstadt.
- **Philip Wadler, Professor (WP4).** Professor of Theoretical Computer Science at the University of Edinburgh and a globally recognised leader in programming languages and formal methods, with foundational contributions to functional programming.
- **Evangelos Markakis, Senior Research Fellow (WP5).** Professor at Athens University of Economics and Business, specializing in game theory, optimisation, and incentive design.
- **Qiang Tang, Associate Professor (WP5).** Associate Professor at the University of Sydney specialising in blockchain and applied cryptography, with publications across

top security and cryptography venues.

- **Paolo Penna, Senior Research Fellow (WP6).** Leading expert in tokenomics, mechanism design, and algorithmic game theory, with 100+ publications and prior senior research roles at ETH Zurich.
- **Sanjam Garg, Professor (WP6).** Leading cryptographer at the University of California, Berkeley, specialising in public-key cryptography, zero-knowledge proofs, and functional encryption.
- **Agustín Garassino, Research & Engineering Lead (WP6).** Partner at Eryx and part-time professor at Universidad Torcuato Di Tella, specialising in blockchain infrastructure and systems development.
- **Mirjam Wester, Project Management Lead (WP7).** Senior Research Manager at Input Output with 20+ years' experience, combining a strong publication record (60+ papers) with extensive expertise in program management and collaborative research delivery.
- **Roger Willis, Head of Product, Cardano (WP7).** Head of Product at Input Output, leading product strategy and execution for Cardano, with responsibility for KPIs including TVL and transaction growth.
- **Fergie Miller, Director (WP7).** Director of Research Partnerships at Input Output, leading research commercialisation, the Cardano Vision R&D program, and the organisation's Grants Office, with 25+ years' experience in deep-tech venture building and innovation.

3. Budget

The total proposed budget of ₦32.917M (\$7.9M) funds all IOR activities across IOR (research), ARC (technology), and CBU (product), supporting a combined 36 FTEs.

This equates to an average cost of approximately \$219.5k per FTE per year (or ~\$946 per day based on 232 working days), inclusive of all associated costs. These include personnel, equipment, software licences, cloud and server infrastructure, subcontracting (eg, academic partners and specialist engineers), as well as program management, travel, events, ecosystem engagement, and dissemination.

IOR (Scientific Research) – \$4.2M | 19.5 FTEs

The research budget of approximately \$4.2M supports 19.5 FTEs across 15 programs. Each workstream requires a multidisciplinary mix of expertise, including distributed systems, consensus, protocol security, applied cryptography, game theory, and formal methods.

Teams typically range from 2–4 contributors per stream, incorporating both internal researchers and external academic partners.

Typical roles include:

- Chief Scientist: Oversees research strategy, scientific direction, and academic excellence
- Professors: Domain-leading experts contributing to advanced research and publication
- Senior Research Fellows: Lead research streams, mentor teams, and drive outputs
- Research Fellows: Deliver specialised research and contribute to publications
- Researchers (PhDs / Associates): Support research delivery and develop domain expertise
- Research & Engineering Specialists: Provide formal methods, cryptographic, and software support

This budget also includes subcontracting and strategic academic partnerships, forming part of IOR’s global research network.

ARC (Technology Innovation) – \$3.7M | 17.5 FTEs

The ARC budget of \$3.7M supports 17.5 FTEs focused on technology validation and implementation readiness. Each workstream requires targeted technical expertise, with teams typically ranging from 1–3 FTEs, depending on scope and complexity. Core roles include:

- Product / Developer Lead: Drives use-case definition, developer engagement, and product alignment
- Technical Architect: Defines system architecture and target implementation environments
- Prototyping Engineers: Develop prototypes, simulations, and validation models
- Applied Cryptographers: Design and benchmark cryptographic components
- Formal Methods Engineers: Develop formal specifications, models, and verification frameworks

ARC acts as the bridge between research and engineering, ensuring outputs are feasible, validated, and implementation-ready.

The budget is structured to support a fully integrated innovation pipeline, combining research excellence (IOR) with applied validation (ARC) and downstream product alignment (CBU). This ensures that investment is directed not only toward knowledge creation, but toward deployable outcomes and measurable ecosystem impact.

Work Package	IOR	ARC	Total FTE
WP1 Trust	5	4.6	8.3

WP2 Scalability	4.5	7.4	9.9
WP3 DevEx	3	2.7	5.7
WP4 Adoption	2	1.6	3.6
WP5 Economics	2.5	1.2	3.7
WP6 Governance	2.5	0	2.5
WP7 Delivery	0	0	0
Total	19.5	17.5	36

IOR proposes that funds are released in four equal tranches of 25%, subject to achievement of the following funding milestones:

- Execution of the Services Agreement
- Submission of the mid-year interim report
- Delivery of the Q3 Cardano R&D session & report
- Submission of the year-end final report

This will be supported by ongoing community communication, dissemination, and stakeholder engagement activities as outlined in Work Package 7.

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5. Appendix

Treasury Governance & Compliance

Contract Management

A written off-chain Legal Contract will be created between Input Output and the Cardano Development Holdings (CDH), as mandated by the Constitution, and will be administered by Intersect. This will include details of the project delivery schedule and dispute resolution.

Project Delivery

All milestones, acceptance criteria, payment amounts and expected delivery dates will be agreed between the Input Output and Intersect, acting on behalf of the CDH. Input Output will deliver according to the agreed-upon project schedule within the Legal Contract, of which the necessary information will be made public via the budget management platform via transaction metadata.

Defined by the milestones within a Legal Contract, Input Output will submit and attest milestone acceptance to the community, Intersect or 3rd Party Assurer.

Project progress will be monitored via Intersect's delivery assurance function which will be communicated to the community.

Acceptance of the work will be supported by a 3rd Party Assurer, who will be responsible for reviewing and signing off the work completed at each project milestone against the corresponding milestone deliverables detailed within the Legal Contract. This work is funded from a portion of this treasury withdrawal.

Auditable Accounts & Fund Delegation

Budget Management Tooling

To administrate treasury funds on-chain, Intersect will utilize the treasury management smart contract framework developed by Sundae Labs. The smart contracts have been extensively tested including audits from TxPipe and MLabs.

Final mainnet validation test can be seen via the Disburse action within transaction: 0f591dc544ae14102dbb4a74d5311a6acffc1772b163d8b7a9656b9525950b17

This withdrawal will utilise Intersect's 2025 treasury reserve contract with address being: stake17xzc8pt7fgf0lc0x7eq6z7z6puhsxmzktna7dluahjrj6g6ghh5qjr

Funds will later be migrated to a 2026 treasury reserve contract once established.

Budget Management Specifics

Intersect will utilize a single Treasury Reserve Smart Contract (TRSC), with many Project-Specific Smart Contracts (PSSC), managed by Intersect. Intersect's management consists of three 'admin' and two Intersect 'leadership' roles. An Oversight Committee consisting of five external, independent third-party entities will provide checks and balances on Intersect, and safeguard against errors and unilateral control. The administration of both TRSC and PSSCs will be managed by Intersect, with external oversight on certain actions from the Oversight Committee.

The 2025 TRSC Oversight Committee consists of Sundae Labs, Cardano Foundation, Dquadrant, Xerberus and NMKR. Their role is to independently verify key administrative actions using on-chain logic, ensuring accuracy and consistency without exercising discretion over governance decisions.

For all details on Intersect's configuration please see the Smart Contract Guide on the knowledgebase.

The high level permissions are as follows:

- TRSC Fund and PSSC Modify
 - Two of the three Intersect admins, two of the five trusted entities and one of the two Intersect leadership sign-off must authorize
- TRSC Disperse
 - Two of three Intersect admins, three of five trusted entities and two of two Intersect leadership sign-off must authorize
- TRSC Pause and Resume
 - Two of three Intersect admins, and one of two Intersect leadership sign-off must authorize
- TRSC Sweep
 - One of three Intersect admins, and one of two Intersect leadership sign-off must authorize
- TRSC Reorganize
 - Two of three Intersect admins and three of five trusted entities must authorize

Processes

Upon enactment of this governance action, funding for this project will be directed into the TRSC's stake account. All instances of TRSC and PSSC can not be staked with a SPO and will be delegated to the auto-abstain predefined DRep. From here funds will be withdrawn into a UTxO remaining at the TRSC.

When a 2026 TRSC is established, the funding for this project will be migrated via the 'disburse' action.

When the Legal contract is prepared and Input Output is ready, funding for this project will be transferred using the Fund action to a PSSC. All milestones will be outlined within the metadata.

A dashboard will be available for the community to audit the TRSC or PSSC and track metrics related to this withdrawn ada as well as being immutably verifiable on chain.

Funding Denomination

All amounts in this proposal are denominated in ada (₳). The total Treasury ask is ₳32.916,667. USD figures (\$7,900,000) are provided for reference only, based on an ADA/USD rate of 0.24.

Refund Conditions

All funds not disbursed by the end of the delivery period will be returned to the Cardano Treasury. A final reconciliation will be published as part of the oversight reporting cycle. In the event of partial delivery or scope reduction, unspent funds associated with cancelled or reduced deliverables will be returned proportionally.

Prior Treasury Receipts

IO and its affiliated entities has been accountable for delivery of work funded by the Cardano Treasury. The total funds allocated has been ₳130,708,860 across a number of projects within Treasury Smart Contract, to date IOG has withdrawn ₳78,459,777.

Workstream	Ada received	% of allocation	Corresponding Governance Action
Blockfrost	₳1,137,500	88%	8ad3d454f3496a35cb0d07b0fd32f687f66338b7d60e787fc0a22939e5d8833e#2
Catalyst	₳3,095,400	60% **	8ad3d454f3496a35cb0d07b0fd32f687f66338b7d60e787fc0a22939e5d8833e#23
IOE	₳47,159,487	49%	8ad3d454f3496a35cb0d07b0fd32f687f66338b7d60e787fc0a22939e5d8833e#1
IOR	₳26,840,000	100%	8ad3d454f3496a35cb0d07b0fd32f687f66338b7d60e787fc0a22939e5d8833e#32
Governance	₳227,390	38%	8ad3d454f3496a35cb0d07b0fd32f687f66338b7d60e787fc0a22939e5d8833e#22

**Note: for Catalyst this only reflects the workstream that focuses on the Hermes Infrastructure and UX/UI improvements, not the execution and operation of Funds 14-16. Per [Info Action](#) this is in the process of transitioning to Cardano Foundation.

Net Change Limit Compliance

The requested amount does not at time of submission, on its own or in aggregate, breach the applicable 350M Net Change Limit covering Epoch 613 to Epoch 713.

In accordance with the guardrail TREASURY-02a, this withdrawal does not exceed the NCL at the moment of submission.

Audit & Oversight

Audit and oversight costs are included within the overhead applied to this proposal. The Intersect administration fee covers administrative oversight and is reflected within the cost of this proposal. Independent oversight will be provided through Intersect and technically capable third-party, including reporting obligations and milestone-based disbursement controls.

Standardized Format & Immutable Hosting

Upon finalization, this proposal will be hosted on IPFS in an immutable format. The blake2b-256 hash of the document will be provided for on-chain reference and verification.